

PAPER_1647_UHECR_E_MAX_7E20_EV

Daniel T. Murphy

PAPER_1647 — UHECR Maximum Energy = $E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^{\blacksquare} = 7 \times 10^{2\blacksquare} \text{ eV}$ (~70 EeV GZK)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: High-Energy Astro

Date: June 18, 2026

Status: CLOSED — 0.5% closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1322 (PAPER_13xx broader corpus) gives:

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$E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^{\blacksquare} = 7 \times 10^{2\blacksquare} \text{ eV} \text{ (~70 EeV GZK)}$

``

Status: 0.5%.

UQFF Closed Identity

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$E_{\text{max}} = K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^{\blacksquare} = 7 \times 10^{2\blacksquare} \text{ eV} \text{ (~70 EeV GZK)} \text{ } 0.5\%$

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical high-energy astro measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1322

- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)

- Calculator dispatch: `calculate_paradox({"paradox": "uhecr_e_max_7e20_ev"})`

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